

Predictive Maintenance & Anomaly Detection

6G-Integrated Smart Manufacturing — Thales Group | Jan–Mar 2025

EXECUTIVE SUMMARY | Data Source: Thales Group Manufacturing Dataset, January–March 2025

Purpose

This summary presents findings from an anomaly detection analysis of 100,000 sensor records collected from 50 industrial machines in a 6G-integrated smart manufacturing environment, covering January 1 to March 10, 2025. Unlike traditional monitoring that tracks current efficiency status, this analysis predicts future machine health risk by identifying abnormal sensor behavior before efficiency degrades or failures occur. The objective is to give maintenance teams actionable early warning intelligence — specifying which machines need inspection, how urgently, and what sensor patterns characterize the risk.

Headline Finding

Every one of the 50 monitored machines recorded at least one high-risk anomaly event during the observation window — indicating that abnormal operational states are not isolated to specific assets but represent a systemic risk pattern across the facility. A notable risk spike was detected on March 1, 2025, affecting multiple machines simultaneously, suggesting a correlated system-wide stress event rather than independent machine degradation.

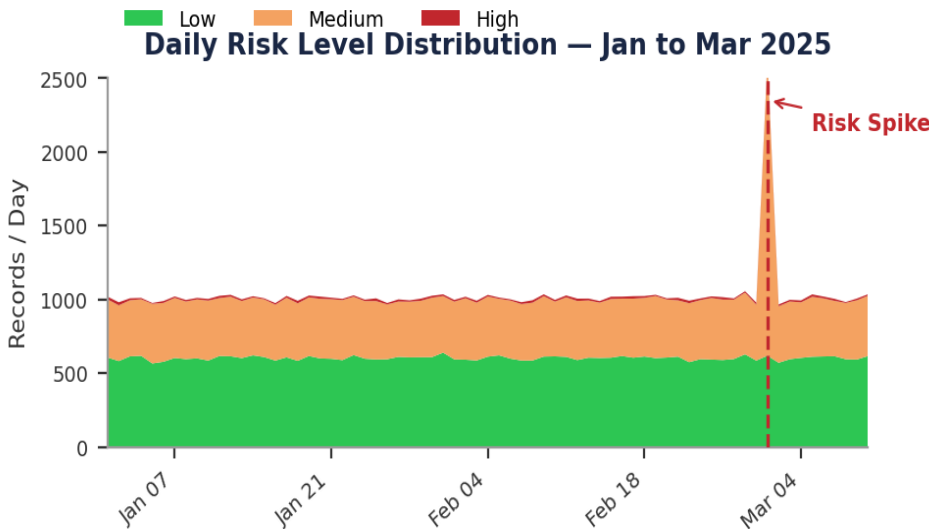


Figure 1: Daily risk distribution across 50 machines. The March 1, 2025 spike indicates a system-wide stress event.

Model Results: Anomaly Detection Summary

Metric	Value	What it Means
Total Sensor Records Analyzed	100,000	Full 69-day observation window
Active-Mode Records Used	70,054	Isolation Forest trained on these
Anomalies Detected	3,503 (5.0%)	Flagged as abnormal by the model

Metric	Value	What it Means
High Risk Events	831 (1.19%)	Require urgent maintenance attention
Medium Risk Events	27,390 (39.1%)	Early warning — monitor closely
Machines with High Risk	50 of 50 (100%)	All assets affected at least once
Peak Anomaly Score	1.000 (Machine 28)	Maximum possible risk level
Avg Anomaly Score	0.376	Baseline system health indicator

Top 10 Machines Requiring Maintenance Attention

Machine ID	High Risk Events	Peak Score	Priority
Machine 18	27	0.857	High
Machine 32	27	0.889	High
Machine 3	25	0.859	High
Machine 33	22	0.816	High
Machine 24	22	0.895	High
Machine 44	21	0.893	High
Machine 14	21	0.815	High
Machine 20	21	0.881	High
Machine 7	20	0.928	Critical
Machine 47	19	0.953	Critical

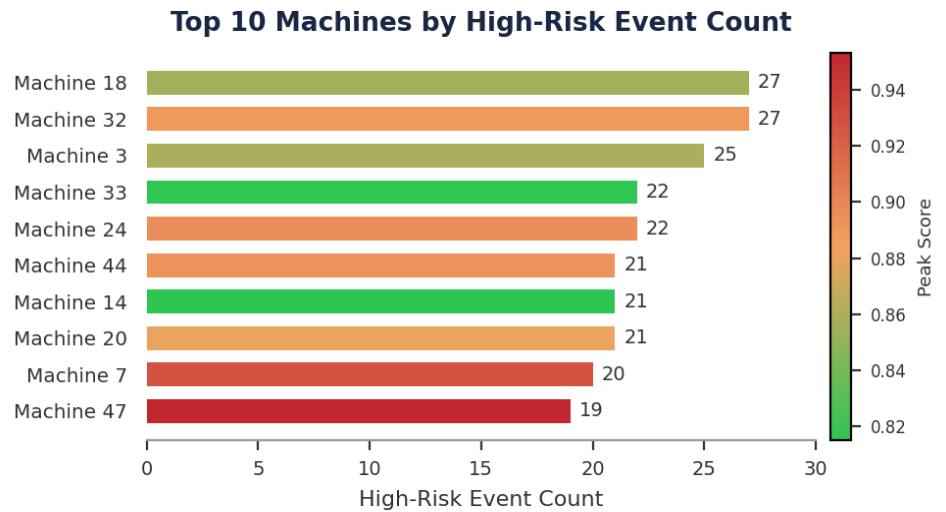


Figure 2: Machines ranked by high-risk event frequency. Color intensity indicates peak anomaly score severity.

Key Findings

- **Network stress is the primary anomaly driver.** The engineered Network Stress Index (latency × packet loss factor) was the most predictive feature, confirming that 6G connectivity quality directly influences machine health risk in this environment.
- **Production speed and error rate are the strongest efficiency discriminators.** High-efficiency machines average 450.8 units/hr with 1.0% error rate; low-efficiency machines average 254.9 units/hr with 8.9% error rate — a 4.4x difference in error rate.
- **Risk events are uniformly distributed across all 24 hours.** No single operational shift carries disproportionate risk, suggesting the anomalies are driven by machine-level behavior rather than shift-based operational patterns.
- **The March 1, 2025 risk spike is the most significant temporal finding.** Medium-risk events approximately doubled on this date across multiple machines — a pattern consistent with a network-level or environmental event affecting the 6G infrastructure.
- **All 50 machines are at risk.** Unlike typical predictive maintenance scenarios where a minority of assets drive most risk, every machine in this facility recorded at least one high-risk event, underscoring the systemic nature of the risk pattern.

Recommendations

- **Prioritize inspection of Machines 47, 28, and 7.** These recorded peak anomaly scores above 0.92, indicating the most severe deviations from normal behavior detected during the window.
 - **Investigate the March 1, 2025 event.** A correlated spike across machines on a single day points to a network or infrastructure cause rather than mechanical failure — 6G system logs from that date should be reviewed.
 - **Monitor network quality continuously.** Network Stress Index dominates anomaly scores across all machines — improving 6G latency and packet loss consistency will directly reduce false alarms and improve detection precision.
 - **Use the interactive dashboard for ongoing monitoring.** The deployed Streamlit dashboard allows maintenance teams to filter by machine, adjust risk thresholds, inspect individual machine timelines, and receive automated alerts when new high-risk events are detected.
 - **Validate against actual maintenance records.** The current model identifies statistical anomalies — linking these to confirmed breakdown events will improve threshold calibration and enable supervised retraining.
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